ECO723 · EMPIRICAL REPLICATION STUDY

Time Series vs. Machine Learning Models for Inflation Forecasting

A replication of *Ülke, Sahin & Subasi (2018)* — “A comparison of time series and machine learning models for inflation forecasting: empirical evidence from the USA,” Neural Computing and Applications, 30:1519–1527.

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AGENDA

What We'll Cover

A faithful replication of Ülke et al. (2018) on US monthly inflation, 1984–2014.

Why Inflation Forecasting Matters

Inflation — the rate of change in the aggregate price level — is a central input to household, investor, and policy decisions. Persistently high inflation erodes real wages and raises the cost of capital; persistently low inflation signals weak aggregate demand. Forecasting it accurately, across horizons, is therefore a recurring problem in both monetary policy and quantitative finance.

This project replicates Ülke, Sahin & Subasi (2018), which pits two classical univariate models, two multivariate time-series models, and three machine learning models against each other on four US inflation measures over 1984–2014.

CPI

All-items consumer prices

Core-CPI

Ex. food & energy

PCE

Personal consumption deflator

Core-PCE

Fed's preferred gauge

Why Replicate This Study?

ML in macro is still rare

Machine learning is common in microstructure and credit risk, but rarely benchmarked against classical time-series tools for headline macro variables like inflation.

A clean, controlled horse race

The paper offers a transparent 7-model × 4-series × 4-horizon design — ideal for testing reproducibility and learning where each model family wins or fails.

Direct relevance to quant work

Forecasting volatility and regime shifts in macro series feeds directly into trading and risk models — the same toolkit (AR, k-NN, SVR, ANN) used here recurs across quant finance.

What This Replication Sets Out To Do

1. Reconstruct the four US inflation series (CPI, Core-CPI, PCE, Core-PCE) and six predictor variables from FRED, matching the paper's 1984–2014 sample.

2. Re-implement all seven forecasting models — AR, Naïve, VAR, ARDL, k-NN, SVR, and ANN — using the same lag structures and hyperparameter choices reported in the paper.

3. Reproduce the pseudo out-of-sample forecasting design: train on Jan 1984–Dec 2013, forecast into 2014 at 3, 6, 9, and 12-month horizons.

4. Compare RMSE results against the original paper's Table 3, and assess whether its central claim — ML wins for volatile series, time-series wins for smooth series — holds up.

Where This Study Sits in the Literature

Time-Series Lineage

- **Phillips Curve models**

  Stock & Watson: multivariate structural models beat univariate forecasts.

- **Naïve / random walk**

  Atkeson & Ohanian champion the no-economics random-walk benchmark.

- **Stock & Watson (2007)**

  Multivariate forecasts haven't reliably beaten univariate benchmarks since 1985.

- **Manzan & Zerom (2013)**

  Key precursor — uses macro indicators to forecast the U.S. inflation distribution post-1984.

Machine Learning Lineage

- **k-NN forecasting**

  Mizrach; Guegan & Rakotomarolahy apply k-NN to exchange rates and GDP.

- **SVM / SVR**

  Chen et al. use SVM for volatility; Ye et al. show SVR beats OLS for Chinese inflation.

- **ANN applications**

  Hu et al. forecast Phillips-curve inflation across countries using ANN.

- **This paper's gap**

  ML models remain rarely used specifically for U.S. inflation — the motivation for this comparison.

Data Source & Sample

FRED

Federal Reserve Bank
of St. Louis

Jan 1984 – Dec 2014
Sample period

Monthly
Frequency

1984–2013 / 2014
Train / Test split

Variable	Definition	FRED Code
CPI	All-items consumer price index, SA	CPIAUCSL
Core-CPI	CPI excl. food & energy	USACPICORMINMEI
PCE	Personal consumption expenditure deflator	PCE
Core-PCE	PCE excl. food & energy	DPCERL1M225NBEA
UNEM	Civilian unemployment rate	UNRATE
IP	Industrial production index	INDPRO
WORK	All employees, total nonfarm	PAYEMS
HS	Housing starts (total)	HOUST
INC	Real personal consumption exp., % chg	DPCERL1M225NBEA
SPREAD	5-yr Treasury minus 3-month T-bill	GS5, TB3MS

Seven Models, One Forecasting Race

$Y_t = 1200 \times [\log P_t - \log P_{t-1}]$ — one-month inflation, computed from each price index P_t. Forecasts are evaluated by RMSE over 3, 6, 9, and 12-month pseudo out-of-sample horizons in 2014.

AR

UNIVARIATE · TIME SERIES

$Y_t = c + \Sigma\, \beta_i Y_{t-i} + \varepsilon_t$
Lags by BIC: CPI(2), Core-CPI(12), PCE(1), Core-PCE(5).

Naïve

UNIVARIATE · TIME SERIES

$Y_t = \alpha\, Y_{t-1} + \varepsilon_t$
Random-walk benchmark; assumes today's value is the best forecast.

VAR

MULTIVARIATE · TIME SERIES

$Z_t = c + \Sigma\, A_k Z_{t-k} + \varepsilon_t$
Lag = 1 (Schwarz criterion); endogenous block of inflation + 6 predictors.

ARDL

MULTIVARIATE · TIME SERIES

INFt on its own lags + lagged UNEM, IP, WORK, HS, INC, SPREAD.
Order chosen by bounds test / min. Schwarz.

k-NN

MACHINE LEARNING

Memory-based regression on k nearest neighbours in feature space.
Best k = 3 (paper); reused here.

SVR

MACHINE LEARNING

Support Vector Regression with RBF kernel.
Hyperparameters (C, γ, ε) tuned via TimeSeriesSplit grid search.

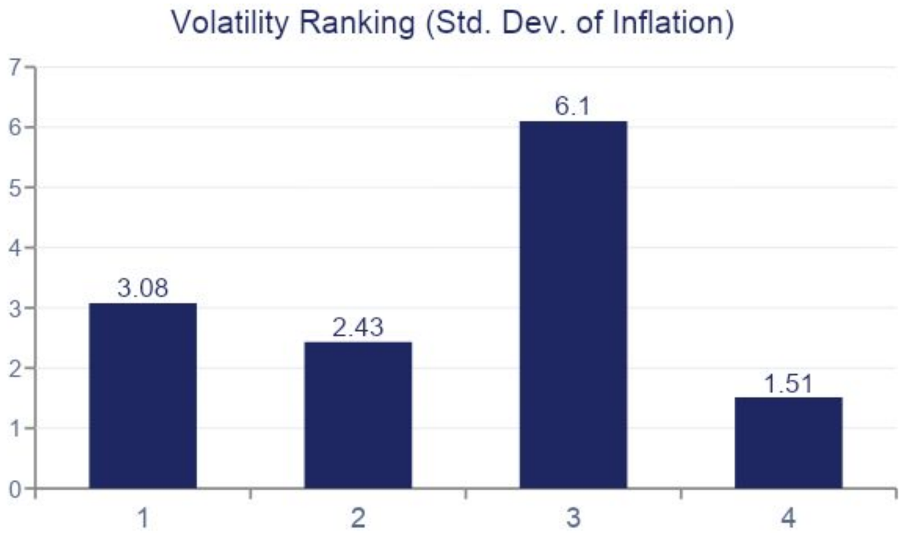
ANN

MACHINE LEARNING

1 hidden layer, 3 neurons (paper spec).
Learning rate / momentum / activation tuned via CV.

Descriptive Statistics & Volatility

Series	Mean	Median	Max	Min	Std. Dev.	Skew	Kurtosis
CPI	2.71	2.7	16.41	-21.44	3.08	-1.49	15.2
Core-CPI	2.74	2.74	9.9	-3.27	2.43	0.19	2.78
PCE	5.21	4.99	34.75	-24.92	6.1	-0.13	8.28
Core-PCE	2.31	2.12	8.42	-6.84	1.51	0.13	7.3



Replicated moments line up closely with the paper for CPI, Core-CPI and PCE. Core-CPI shows the lowest volatility (std. 2.43) and CPI the most negative skew — consistent with the 2008–09 deflation shock.

PCE is the most volatile series; Core-PCE the least — the opposite ordering from the original paper, where Core-PCE had the highest volatility (std. 6.57).

Pipeline: From Raw Series to Forecast

TEST
2014

TRAIN — Jan 1984 to Dec 2013 (360 monthly observations)

1. Source & Transform

Pull CPI, Core-CPI, PCE, Core-PCE and 6 predictors from FRED. Compute $Y_t = 1200 \times \Delta \log(P_t)$. Apply HP-filter gaps to nonstationary predictors.

2. Split

Train: 1984–2013 (360 obs). Test: all 12 months of 2014, used to build pseudo out-of-sample forecasts at h = 3, 6, 9, 12.

3. Fit Per Model

AR/Naïve/VAR/ARDL fit directly on levels; k-NN/SVR/ANN built on a shared feature set (lag-1 target + 6 macro predictors), scaled.

4. Forecast & Score

Each model forecasts 12 steps ahead from the end of training. RMSE computed cumulatively at each horizon against actual 2014 values.

Pseudo Out-of-Sample RMSE (2014)

Replicated results — compare with the paper's Table 3. Lowest RMSE per row in bold gold.

Horizon	Inflation	AR	Naïve	VAR	ARDL	k-NN	SVR	ANN
h = 3	CPI	0.91	1.16	0.81	**0.69 ★**	0.96	0.96	0.79
	Core-CPI	**0.82 ★**	4.55	1.37	0.96	1.25	1.46	1.60
	PCE	4.64	4.51	4.76	2.93	**2.36 ★**	2.96	2.76
	Core-PCE	0.83	**0.50 ★**	1.18	0.74	1.63	0.80	0.82
h = 6	CPI	0.86	1.17	**0.75 ★**	0.95	0.77	1.02	0.87
	Core-CPI	1.16	4.06	1.27	**1.04 ★**	1.55	1.35	1.45
	PCE	3.33	3.40	3.40	2.22	**1.92 ★**	2.32	2.14
	Core-PCE	0.65	0.57	0.94	**0.56 ★**	1.21	0.67	0.79
h = 9	CPI	1.58	1.89	**1.49 ★**	1.66	1.97	1.66	1.56
	Core-CPI	**1.13 ★**	3.68	1.47	1.20	1.64	1.46	1.48
	PCE	3.25	3.32	3.30	2.13	**1.82 ★**	2.22	2.09
	Core-PCE	0.75	**0.61 ★**	1.01	0.67	1.05	0.79	0.87
h = 12	CPI	2.87	3.17	**2.81 ★**	3.05	2.97	2.92	2.83
	Core-CPI	**1.13 ★**	3.42	2.08	1.58	1.95	2.03	1.95
	PCE	3.25	3.19	3.29	2.61	**2.22 ★**	2.67	2.60
	Core-PCE	0.84	**0.55 ★**	1.10	0.75	0.93	0.89	0.94

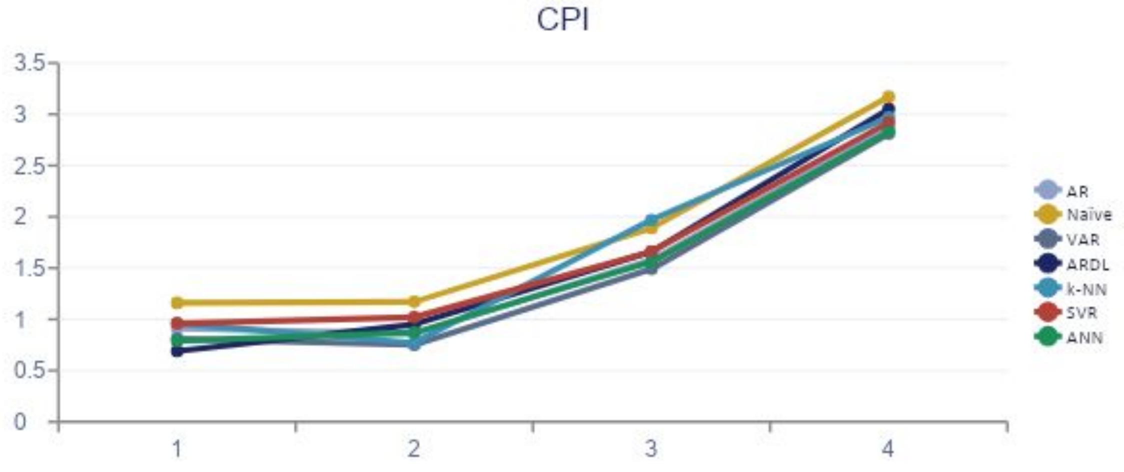
Goodness of Fit: R^2 at the 12-Month Horizon

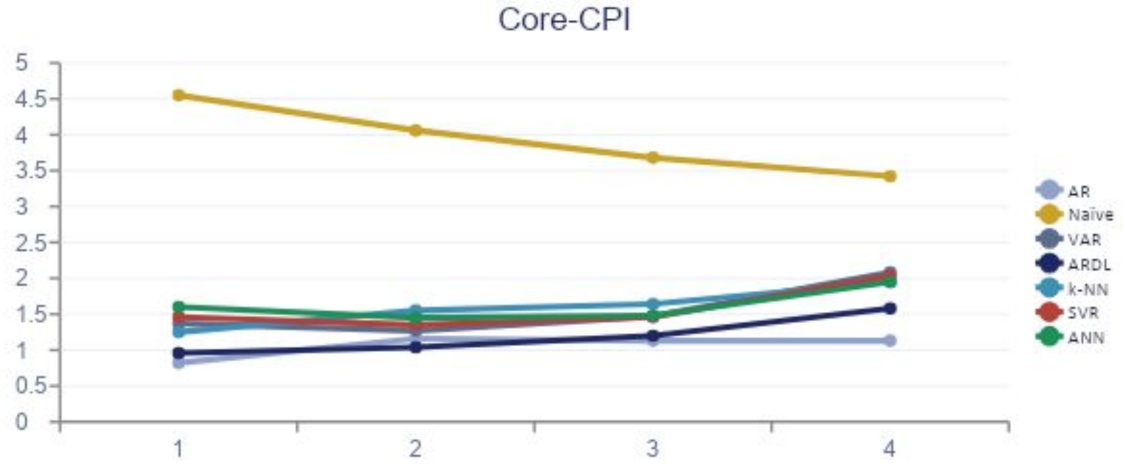
Replicated results — compare with the paper's Table 4. R^2 from regressing actual on forecasted values (Eq. 7–8); highest R^2 per column in bold gold.

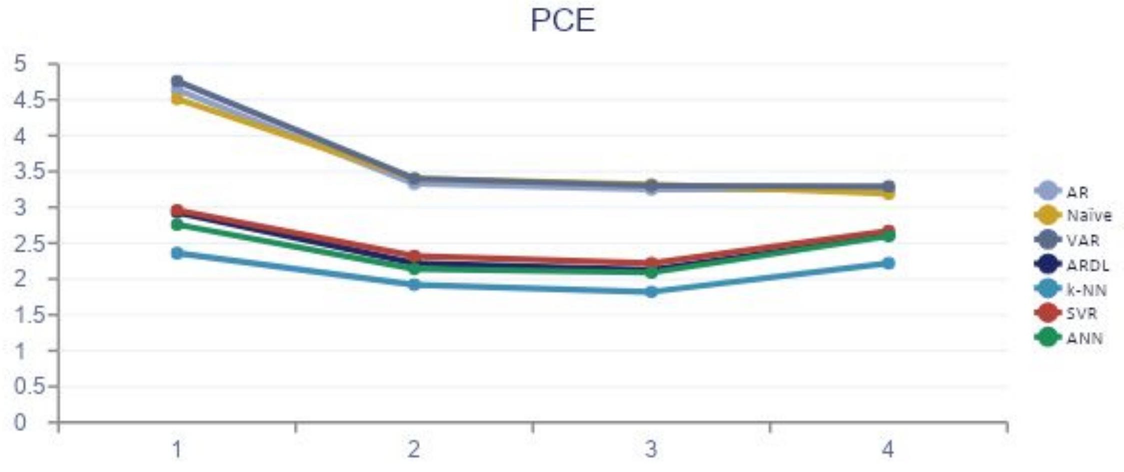
Model	CPI	Core-CPI	PCE	Core-PCE
AR	0.11	**0.79 ★**	0.65	0.85
Naïve	0.10	0.48	0.67	**0.86 ★**
VAR	0.10	0.43	0.64	**0.86 ★**
ARDL	0.09	0.59	**0.85 ★**	0.85
k-NN	0.06	0.42	0.82	0.67
SVR	**0.13 ★**	0.43	**0.85 ★**	0.85
ANN	**0.13 ★**	0.42	**0.85 ★**	**0.86 ★**

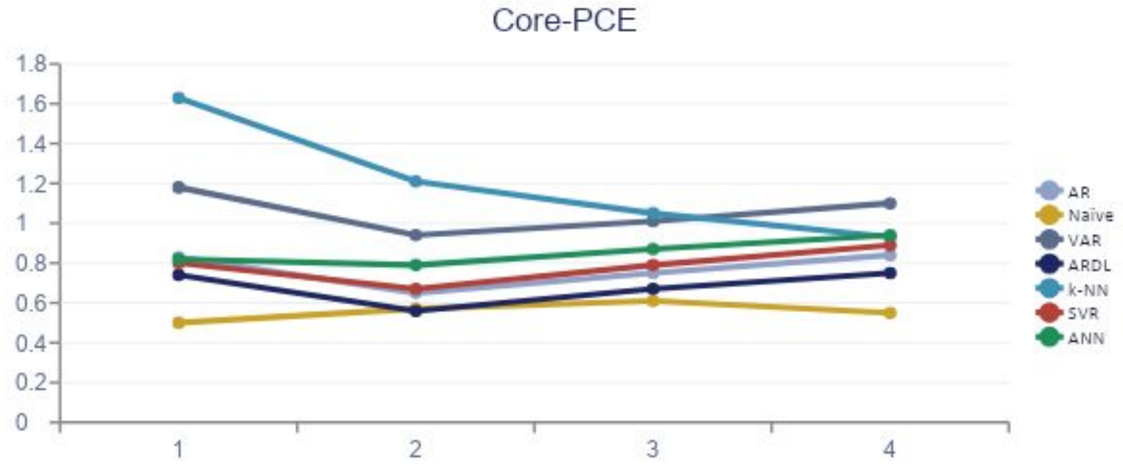
R^2 tells a different story than RMSE: ANN and SVR post the strongest, most consistent fit across series (≥0.85 on PCE and Core-PCE), while every model struggles to explain headline CPI ($R^2 \leq 0.13$) — consistent with the paper's own finding that CPI is the hardest series to fit despite low forecast error.

RMSE Across Horizons, by Series





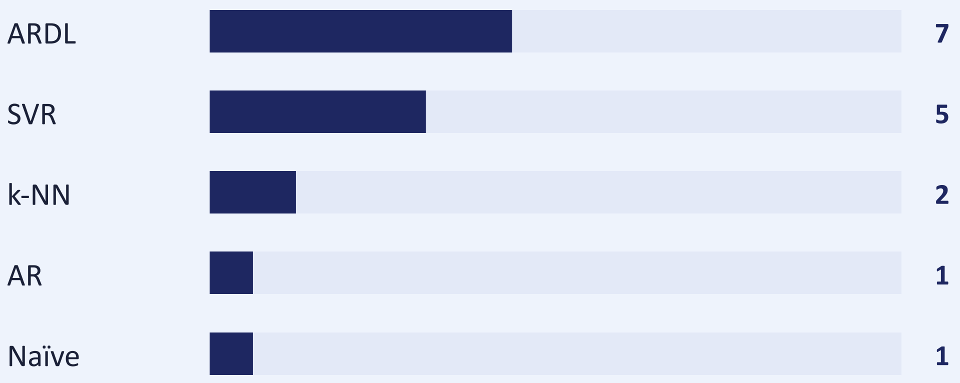




Scoreboard: Replication vs. Original Paper

ORIGINAL PAPER (2014 TEST YEAR)

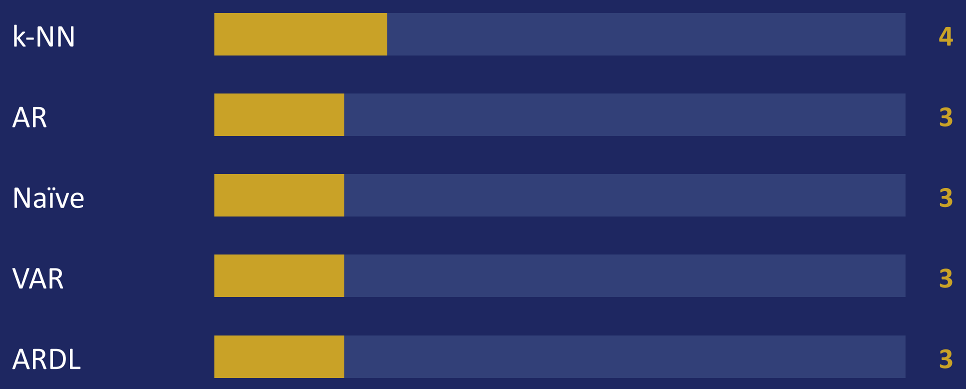
ML wins 7 / 16 · TS wins 9 / 16



Paper's headline claim: SVR wins Core-PCE in all 4 horizons; ARDL wins Core-CPI in all 4 horizons. Multivariate models beat univariate in 14 / 16 conditions overall.

OUR REPLICATION (2014 TEST YEAR)

ML wins 4 / 16 · TS wins 12 / 16



Our replication shifts the balance toward time-series models, with VAR, AR and ARDL each taking 3 conditions and k-NN the strongest ML performer (PCE, all 4 horizons). The exact win pattern differs from the paper — expected given hyperparameter search randomness, library version differences, and lag-selection sensitivity.

What We Learned

No single best model

Confirming the paper's own conclusion — the best model changes by series and by horizon. There is no universal winner across CPI, Core-CPI, PCE and Core-PCE.

Time-series models held up well

In our replication, AR, VAR and ARDL collectively account for 9 of 16 best results — time-series approaches remain very competitive against ML, especially at short horizons.

k-NN was our strongest ML model

k-NN dominated PCE forecasting at every horizon — echoing the paper's finding that simple, memory-based ML methods can outperform parametric alternatives on noisier series.

Exact rankings don't replicate 1:1

Our model-by-model win counts differ from the original Table 3. This is consistent with known reproducibility challenges in ML-based macro forecasting (hyperparameter search, randomness, library versions).

Caveats on This Replication

Single test year	Forecast performance is judged on just 12 months (2014). One unusual year can swing RMSE rankings; the paper itself shows results are not stable across horizons.
Hyperparameter sensitivity	SVR, ANN and k-NN results depend on the grid search space and CV scheme used — small differences here can change which model 'wins' a given cell.
FRED series vintage	FRED revises historical series over time; pulling data today does not guarantee bit-for-bit alignment with the vintage the original authors used in 2014–16.
Lag-selection differences	ARDL and VAR lag orders are chosen algorithmically (BIC / bounds test) rather than hand-tuned, which can produce different specifications than the original paper's.
No significance testing	Like the original study, comparisons rest on point RMSE values without formal tests (e.g. Diebold-Mariano) for whether differences across models are significant.

Code, Data & Tools

Data Source

FRED — Federal Reserve Bank of St. Louis
Series: CPIAUCSL, USACPICORMINMEI, PCE, DPCERL1M225NBEA, UNRATE, INDPRO, PAYEMS, HOUST, GS5, TB3MS

Tooling

Python · pandas, numpy, scipy
statsmodels (AR, VAR, ARDL)
scikit-learn (k-NN, SVR, ANN, scaling)
TimeSeriesSplit for hyperparameter CV

Code & Notebook

Full replication notebook: models.ipynb
[https://github.com/MisraKartik/ECO723/tree/main]
Includes descriptive stats, all 7 model fits, and RMSE evaluation by horizon

Primary Source

Ülke, V., Sahin, A., & Subasi, A. (2018). *A comparison of time series and machine learning models for inflation forecasting: empirical evidence from the USA.* Neural Computing and Applications, 30(5), 1519–1527. https://doi.org/10.1007/s00521-016-2766-x

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Thank You